RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2022-23) -- SEMESTER 3

COURSE: DISCRETE STRUCTURES & THEORY OF LOGIC (CS-303)

Time Allowed: 3 Hours Maximum Marks: 100 Marks Paper Code: CS-303

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Define equivalence relation and equivalence classes with an example.

Q1. (b) What is partially ordered set (POSET)? Define Hasse diagram.

Q1. (c) State Pigeonhole Principle and its generalized form.

Q1. (d) Define group, abelian group, and semi-group with algebraic axioms.

Q1. (e) What is Lagrange theorem in finite group theory?

Q1. (f) Define lattice and bounded lattice with meet and join operations.

Q1. (g) Check if propositional formula (p -> q) <-> (~p v q) is a tautology.

Q1. (h) Define Eulerian graph and Hamiltonian graph with necessary and sufficient conditions.

Q1. (i) What is planar graph? State Euler formula relating vertices, edges, and faces.

Q1. (j) Solve recurrence relation a_n = 5 a_(n-1) - 6 a_(n-2) with a_0 = 1, a_1 = 4.

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Show that (Z_7 - {0}, *_7) forms an abelian group under multiplication modulo 7.

Q2. (b) Draw Hasse diagram for divisibility relation on D_30 = {1, 2, 3, 5, 6, 10, 15, 30}. Is it a Boolean algebra?

Q2. (c) Prove by mathematical induction that 1^2 + 2^2 + 3^2 + ... + n^2 = n(n+1)(2n+1)/6 for all n >= 1.

Q2. (d) Find the generating function and solve the recurrence relation a_n - 7 a_(n-1) + 10 a_(n-2) = 0 with a_0 = 1, a_1 = 2.

Q2. (e) Prove that a connected planar graph with v vertices and e edges satisfies e <= 3v - 6 if v >= 3.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Prove that intersection of two equivalence relations is also an equivalence relation. What about union? [10 Marks]

-- OR --

(b) Let R be a relation on integers defined by aRb iff a - b is divisible by 5. Prove R is equivalence. [10 Marks]

Q4. (a) State and prove Lagrange theorem for finite groups. Does the converse hold true? [10 Marks]

-- OR --

(b) Explain normal subgroups, quotient groups, and group homomorphism with kernel definition. [10 Marks]

Q5. (a) Define complemented, distributive, and modular lattices with examples and diagrams. [10 Marks]

-- OR --

(b) Express Boolean function f(x, y, z) = (x + y)(y' + z) in complete sum-of-products (SOP) canonical form. [10 Marks]

Q6. (a) Convert formula into Conjunctive Normal Form (CNF): ~((p -> q) ^ (q -> r)). [10 Marks]

-- OR --

(b) Prove validity of argument using rules of inference: p -> q, ~q v r, ~r |- ~p. [10 Marks]

Q7. (a) Prove that every tree with n vertices has exactly n - 1 edges. [10 Marks]

-- OR --

(b) Explain graph isomorphism, chromatic number of a graph, and Four Color Theorem. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****